

# Model-agnostic variable importance for survival outcomes: a simulation study under additive and mixed hazards

## Simulation protocol

### Introduction

We conducted a simulation study to examine the behavior of Wolock’s model-agnostic exclusion variable importance measure (VIM) under a range of survival data-generating mechanisms. The objectives were to evaluate its performance and inferential validity under additive hazards, mixed additive–multiplicative hazards, and proportional hazards settings, including high-dimensional scenarios with rare binary covariates and heterogeneous effect sizes. We also compared alternative strategies for nuisance function estimation and contrasted exclusion-based importance with permutation-based importance across all scenarios.

### Objectives

The simulation study was designed to evaluate the performance, robustness, and inferential validity of Wolock’s model-agnostic exclusion variable importance measure (VIM) across several survival data-generating mechanisms.

First, under an **additive hazards framework**, we assess the accuracy and inferential properties of the VIM and compare different strategies for estimating the conditional survival functions, namely *global survival stacking*, *survival Super Learner*, the *Aalen additive hazards model*, and *random survival forests*.

Second, under **mixed additive and multiplicative hazard structures**, we investigate the behavior of the VIM when both types of effects coexist. In this setting, conditional survival functions are estimated using *global survival stacking*, *survival Super Learner*, the *Cox–Aalen model*, and *random survival forests*.

Third, we assess the sensitivity of the VIM when covariates correspond to **rare alterations** with **heterogeneous effect sizes**, ranging from weak to strong, under a **proportional hazards model**.

Finally, we compare **exclusion-based variable importance** with **permutation-based variable importance** to examine differences in interpretation, stability, and empirical performance across the considered scenarios.

## Data-generation mechanisms

### Simulation of covariates

We simulated datasets combining binary and continuous covariates for  $i = 1, \dots, n$  individuals.

#### Binary covariates

Binary covariates were generated as correlated Bernoulli variables:

$$X_{ij} \in \{0, 1\}, \quad j = 1, \dots, p_B,$$

where  $X_{ij} = 1$  indicates the presence of binary feature  $j$ .

Each variable had marginal prevalence

$$\pi_j = \Pr(X_{ij} = 1).$$

To induce correlation between variables, a latent Gaussian copula model was used:

$$\mathbf{Z}_i \sim \mathcal{N}(0, \Sigma_B),$$

where  $\Sigma_B$  is a block-structured correlation matrix.

Binary variables were obtained through probit thresholding:

$$X_{ij} = \mathbb{I}(Z_{ij} < \Phi^{-1}(\pi_j)),$$

where  $\Phi^{-1}$  denotes the quantile function of the standard normal distribution. This construction guarantees

$$\Pr(X_{ij} = 1) = \pi_{j\cdot}.$$

Thus, marginal prevalences are exactly controlled while inducing dependence between binary variables through the latent correlation matrix.

### Continuous covariates

Continuous covariates were simulated from a multivariate normal distribution:

$$\mathbf{W}_i \sim \mathcal{N}(0, \Sigma_C),$$

where  $\Sigma_C \in \mathbb{R}^{p_C \times p_C}$  is a block-correlation matrix.

Variables were partitioned into  $K$  blocks of equal size  $b$ . For block  $k$ , define

$$B_k = \{(k-1)b + 1, \dots, kb\}.$$

The correlation matrix  $\Sigma_C = (\sigma_{jl})$  was defined as

$$\sigma_{jl} = \begin{cases} 1 & \text{if } j = l, \\ \rho_k & \text{if } j \neq l \text{ and } j, l \in B_k, \\ \rho_B & \text{if } j \in B_k, l \in B_m, k \neq m. \end{cases}$$

Equivalently, each diagonal block can be written as

$$\Sigma_{C,k} = (1 - \rho_k)I_b + \rho_k \mathbf{1}_b \mathbf{1}_b^\top,$$

where  $I_b$  denotes the  $b \times b$  identity matrix and  $\mathbf{1}_b$  is a vector of ones of length  $b$ .

Continuous covariates were therefore correlated according to a block dependence structure.

## Simulation of time-to-event outcomes

Let  $T$  denote the event time and  $C$  an independent censoring time. The observed data consisted of  $(Y, \Delta, X, Z)$ , where:

$$Y = \min(T, C), \quad \Delta = \mathbb{I}(T \leq C).$$

Censoring times were generated independently from an exponential distribution to achieve a target censoring proportion.

### Additive Aalen model

Time-to-event outcomes were generated under an additive hazards framework. For each individual  $i = 1, \dots, n$ , with covariate vector  $\mathbf{X}_i$ , the hazard function was defined as

$$\lambda(t | X_i) = \lambda_0(t) + X_i^\top \alpha,$$

where  $\lambda_0(t)$  denotes the baseline hazard function and  $\alpha$  represents the vector of additive hazard effects.

### Cox-Aalen model

To evaluate performance under mixed hazard structures, we additionally generated data from a Cox-Aalen model:

$$\lambda(t | X_i, Z_i) = \lambda_0 \exp(Z_i^\top \beta) + X_i^\top \alpha.$$

where:

- $Z_i$  are covariates with multiplicative effects,
- $\beta$  are multiplicative coefficients,
- $X_i$  are covariates with additive effects,
- $\alpha$  are additive coefficients

## Target estimands and oracle variable importance

We considered two estimators of variable importance, targeting distinct estimands: exclusion importance and permutation importance.

## Model-agnostic framework for exclusion-based variable importance in survival analysis

### Definition

We consider the model-agnostic variable importance measure proposed by Wolock *et al.*, defined through covariate exclusion. For a given prediction time  $t_0$  and covariate  $X_s$ , the VIM quantifies the change in predictive performance when  $X_j$  is removed from the prediction function.

We defined a nonparametric variable importance measure (VIM) based on the predictive value of a variable set within a flexible survival prediction model fitted with **survML**. Let  $V(\cdot)$  denote a time-dependent predictive performance functional, evaluated on predictions. The variable importance measure for covariate  $X_j$  at time  $t_0$ ,  $\psi_j(t_0)$ , was defined as the difference in predictive performance between a model fit using the full feature set,  $V_{full}$  and a model fit after excluding the variable or group of variables of interest,  $V_{reduced}$ :

$$\psi_{X_j}(P_0) = V_{full} - V_{reduced}$$

### Estimation of the conditional survival functions

We estimated the conditional survival functions  $F_0$  and  $G_0$  using different strategies, depending on the scenario:

- Aalen model (`timereg::aalen()`, `pec::predictSurvProb()`)
- Cox-Aalen survival model (`timereg::cox.aalen()`, `pec::predictSurvProb()`)
- Random survival forests (`randomForestSRC::rfsrc()`)
- Survival Super Learner (`survivalSL`)
- Global survival stacking (`survML::stackG()`)

### Estimation of the oracle prediction functions

We estimated the full and residual oracle prediction functions for time-horizon VIMs,  $f_0$  and  $f_{0,s}$ , using Super Learner regression.

We used the sample splitting procedure to compute VIM point and standard error estimates, from which we computed nominal 95% CI Wald-type confidence intervals.

## Permutation variable importance in machine learning survival models

### Definition

Permutation VIM measures was calculated using **survex** package using the `model_parts()` function. The importance of  $j$ -th variable is defined as the change in the loss function  $\mathcal{L}$  caused by permutation of this variable in the dataset:

$$\text{PFI}_t(f, \mathbf{X}, j, \mathcal{L}, y) = \frac{1}{B} \sum_{i=1}^B (\mathcal{L}(f, \mathbf{X}, y) - \mathcal{L}(f, \mathbf{X}_i^{*j}, y))$$

where  $\mathcal{L}$  represents the loss function chosen to evaluate model performance,  $\mathbf{X}_i^{*j}$  denotes the  $i$ -th permutation of variable  $j$  within the dataset  $X$ , and  $B$  is the number of different permutations. Permuting a variable is supposed to simulate the loss of information associated with the variable. For `survex`, we used the `model_parts()` function with 10 permutations.

### Model framework

We used the following models to calculate variable importance in the loss function after the variable values permutations:

- Aalen model (`timereg`, `pec`)
- Cox-Aalen survival model (`timereg`, `pec`)
- Random survival forests (`randomForestSRC`)

### Time horizon

Variable importance was evaluated at a fixed time horizon, corresponding to **75th percentiles** of the empirical observed event times (or the Kaplan-Meier estimator to account for censoring). These represent a late-risk horizon. Restricting the analysis to this quantile-based time point reduces computational burden while ensuring that evaluation occurs in regions with a sufficient number of individuals at risk, thereby avoiding instability in the tail of the distribution.

### Performance metrics for variable importance in survival analysis

Variable importance was evaluated using the *Brier score*,  $BS(t)$  to capture both discrimination and calibration. We additionally used the *cumulative/dynamic area under the ROC curve*,  $AUC(t)$  to estimate variable importance measures, which evaluate discrimination independently of calibration.

The ***Brier score*** at time  $t$  is the expected squared difference between the observed survival status at time  $t$  and the predicted survival probability, with inverse probability of censoring weights to account for right censoring.

The ***cumulative/dynamique AUC*** at time  $t$  is defined as the probability that, among a randomly selected pair consisting of one individual who experiences the event before  $t$  and one who survives beyond  $t$ , the predicted risk is higher for the former.

## Cross-validation procedure

To ensure fair comparison across nuisance estimation strategies, all models were evaluated using the same cross-validation framework.

Five-fold cross-validation was employed for out-of-sample performance assessment and to prevent optimistic bias in risk prediction and downstream VIM estimation.

We implemented  $K$ -fold cross-validation with  $K = 5$ . The data were randomly partitioned into five approximately equal folds. For each fold:

1. The model was trained on the  $K - 1$  remaining folds.
2. Predictions of the conditional survival and hazard functions were obtained for individuals in the held-out fold.
3. Performance metrics were computed on the validation fold.

This procedure was repeated so that each observation contributed once to validation risk estimation.

## Oracle variable importance

We generated a large Monte Carlo sample ( $n = 200000$ ) from the true data-generating model to approximate population-level quantities. Using this sample, we computed the oracle predictive performance as well as the oracle VIM measure for each variables.

## Performance evaluation

The following metrics were computed across  $B = 1000$  Monte Carlo replications.

| Metric              | Definition                                                       | Formula                                                             |
|---------------------|------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Bias</b>         | Average deviation between the estimator and the true parameter   | $\frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{(b)} - \theta)$            |
| <b>RMSE</b>         | Root mean squared error measuring overall estimation accuracy    | $\sqrt{\frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{(b)} - \theta)^2}$   |
| <b>Coverage</b>     | Proportion of confidence intervals containing the true parameter | $\frac{1}{B} \sum_{b=1}^B \mathbf{1}(\theta \in CI^{(b)})$          |
| <b>CI width</b>     | Average length of confidence intervals                           | $\frac{1}{B} \sum_{b=1}^B (U^{(b)} - L^{(b)})$                      |
| <b>Type I Error</b> | Probability of rejecting $H_0$ when it is true                   | $\frac{1}{B} \sum_{b=1}^B \mathbf{1}(p^{(b)} < \alpha)$ under $H_0$ |

| Metric                  | Definition                                                  | Formula                                                             |
|-------------------------|-------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Power</b>            | Probability of rejecting $H_0$ when the alternative is true | $\frac{1}{B} \sum_{b=1}^B \mathbf{1}(p^{(b)} < \alpha)$ under $H_1$ |
| <b>Rank correlation</b> | Correlation between true and estimated parameter rankings   |                                                                     |
| <b>Runtime</b>          | Average computational time per replication                  | $\frac{1}{B} \sum_{b=1}^B t^{(b)}$                                  |

where:

- $B$  is the number of Monte Carlo replications,
- $\theta$  is the true parameter,
- $\hat{\theta}^{(b)}$  is the estimate in replication  $b$ ,
- $CI^{(b)} = [L^{(b)}, U^{(b)}]$ ,
- $\alpha$  is the nominal significance level.

## Simulation scenarios

The main scenarios of the simulation study are summarized in Table 2.

Table 2: Summary of the main scenarios

| Scenario<br>n° | Effect type                                                                     | Simulation of<br>covariates                                | Coefficient                                                                                                                                                                                                  | Nuisance<br>function<br>estimation                                                |
|----------------|---------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1              | Additive<br>hazards<br>(Aalen<br>model)                                         | $X_1, X_2, X_3, X_4$<br>$Bernoulli(0.4)$                   | $\alpha_1 = 0.1$ (strong effect)<br>$\alpha_2 = 0.05$ (moderate effect)<br>$\alpha_3 = 0.005$ (weak effect)<br>$\alpha_4 = 0$ (null effect)                                                                  | Aalen model<br>Survival Super<br>Learner<br>Global<br>survival<br>stacking        |
| 2              | Mixed<br>additive<br>and multi-<br>plicative<br>hazards<br>(Cox-Aalen<br>model) | $X_1, X_2, X_3, Z_1, Z_2$<br>$Bernoulli(0.4)$<br>$N(0, 1)$ | $\alpha_1 = 0.1$ (strong effect)<br>$\alpha_2 = 0.05$ (moderate effect)<br>$\alpha_3 = 0$ (null effect)<br>$\beta_1 = 1$ (strong effect)<br>$\beta_2 = 0.5$ (moderate effect)<br>$\beta_3 = 0$ (null effect) | Cox-Aalen<br>model<br>Survival Super<br>Learner<br>Global<br>survival<br>stacking |



## Scenario 1 - Additive hazards

### Objective

This baseline scenario evaluates the robustness of Wolock’s model-agnostic variable importance measure (VIM) under correct model specification. Specifically, we assess whether competing nuisance-function estimators recover the true exclusion VIM when the data-generating mechanism follows an **additive hazards model**.

### Data generating mechanism

Four binary covariates were generated *independently* with prevalences spanning rare to frequent alterations,  $X_j \sim \text{Bernoulli}(0.4)$ .

Events times were simulated under the following time-constant additive hazards model:

$$\lambda(t|x) = \lambda_0(t) + \alpha_1(t)X_1 + \alpha_2(t)X_2 + \alpha_3(t)X_3 + \alpha_4(t)X_4$$

with  $\lambda_0(t) = 0.05$  (constant) and  $\alpha_j(t) = (0.1, 0.05, 0.005, 0)$ .

Independent censoring times were generated as  $C \sim \text{Uniform}(0, c_{\max})$ . The parameter  $c_{\max}$  was calibrated using a large Monte Carlo sample from the data-generating mechanism to achieve an expected censoring proportion of approximately 20%, and the calibrated value was then fixed across all simulation replicates.

### Nuisance functions estimators to compare

We compare the following approaches for estimating the nuisance functions required for VIM computation:

- Aalen additive hazards model (`timereg::aalen`)
- Survival Super Learner (`survivalSL`)
- Global survival stacking (`survML`)

Since the data-generating mechanism follows an additive hazards model, the additive Aalen estimator is correctly specified. The remaining methods introduce varying degrees of model misspecification, allowing us to assess the robustness of the VIM procedure to nuisance-model choice.

## Scenario 2 - Mixed additive and multiplicative effects

### Objective

This scenario evaluate the sensitivity of the exclusion VIM when the true hazard follows a **Cox-Aalen model**.

### Data-generating mechanism

The covariates are generated *independently* as follows:

- $X = (X_1, X_2, X_3, Z_1, Z_2, Z_3)$
- $X_1, X_2, X_3 \sim \mathcal{B}(0.4)$
- $Z_1, Z_2 \sim \mathcal{B}(0.4)$
- $Z_3 \sim \mathcal{N}(0, 1)$

To generate mixed additive and multiplicative effects, event times were simulated under a Cox-Aalen model with constant baseline hazard:  $\lambda_0(t) = 0.05$ . Additive effects were assigned to  $X_1$ ,  $X_2$  and  $X_3$  with coefficients  $\alpha_1(t) = 0.1$ ,  $\alpha_2 = 0.05$  and  $\alpha_3(t) = 0$ , respectively. Multiplicative (proportional hazards) effects were assigned to  $X_1$ ,  $X_2$  and  $X_3$  with coefficients  $\beta_1 = 1$ ,  $\beta_2 = 0.5$  and  $\beta_3 = 0$ .

Independent censoring times were generated as  $C \sim \text{Uniform}(0, c_{max})$ . The parameter  $c_{max}$  was calibrated using a large Monte Carlo sample from the data-generating mechanism to achieve an expected censoring proportion of approximately 20%, and the calibrated value was then fixed across all simulation replicates.

### Nuisance functions estimators to compare

We compare the following approaches for estimating the nuisance functions required for VIM computation:

- Cox-Aalen model (`timereg::cox.aalen`)
- Survival Super Learner (`survivalSL`)
- Global survival stacking (`survML`)

## Additional numerical experiments

### Sensitivity to low prevalence

#### Objective

This scenario aims to evaluate the overall performance of the VIM procedure to *rare alterations* exhibiting *heterogeneous effect sizes (ranging from null to strong)* under *additive* and a mixed additive and multiplicative hazards. More specifically, the objective is to assess whether the VIM approach can:

- detect rare yet highly prognostic alterations,
- discriminate between strong and moderate prognostic signals, and
- correctly identify non-informative (null) variables.

### Data-generating mechanisms

All covariates were generated independently as binary variables:

$$X_j \sim \mathcal{B}(\pi_j)$$

with  $\pi \in \{0.01, 0.1\}$  to reflect genomic rarity.

The event times follows a Aalen or Cox-Aalen model (cf. scenario 1 and 2).

Covariates were partitioned according to both their prevalence and prognostic effect size, resulting in the following combinations :

| Prevalence category | $\pi_j$      | Effect category | $\alpha$ | $\beta$ |
|---------------------|--------------|-----------------|----------|---------|
| Very rare           | $\pi = 0.01$ | Strong          | 0.10     | 1.0     |
| Very rare           | $\pi = 0.01$ | Moderate        | 0.05     | 0.5     |
| Very rare           | $\pi = 0.01$ | Null            | 0.00     | 0.0     |
| Rare                | $\pi = 0.10$ | Strong          | 0.10     | 1.0     |
| Rare                | $\pi = 0.10$ | Moderate        | 0.05     | 0.5     |
| Rare                | $\pi = 0.10$ | Null            | 0.00     | 0.0     |

This design allows us to assess whether the VIM procedure can detect rare but strongly prognostic alterations, distinguish strong from moderate effects, and correctly assign low importance to null variables.

Independent censoring times were generated as  $C \sim \text{Uniform}(0, c_{max})$ . The parameter  $c_{max}$  was calibrated using a large Monte Carlo sample from the data-generating mechanism to achieve an expected censoring proportion of approximately 20%, and the calibrated value was then fixed across all simulation replicates.

### Sensitivity to covariate correlations

For each scenario, we additionally evaluated the sensitivity of the VIM to correlations between covariates. Correlations were introduced between selected covariates while preserving their marginal distributions (binary or continuous). Specifically, we considered both weak and moderate correlation dependence between predictors, with pairwise correlations set to  $\rho = 0.3$  and  $\rho = 0.6$ , respectively. This analysis allows us to assess the robustness of the estimated VIM in the presence of correlated covariates.

## Compare exclusion variable importance to permutation variable importance

### Objective

To examine differences in interpretation, stability, and empirical performance across the considered scenarios.

### Scenarios

- Additive hazards
- Mixed additive and multiplicative hazards

### True values of variable importance for all scenarios

The approximate true values of variable importance based on  $AUC(t)$  and, Brier score ( $BS(t)$ ) under all scenarios considered here are provided in Table 5 and Table 4, respectively.

Table 4: Approximate true values of variable importance based on cumulative/dynamic AUC

| Scenario ID | Variable     | Global survival stacking | Aalen model | Cox-Aalen model |
|-------------|--------------|--------------------------|-------------|-----------------|
| 1           | $\psi_{X_1}$ |                          |             |                 |
| 1           | $\psi_{X_2}$ |                          |             |                 |
| 1           | $\psi_{X_3}$ |                          |             |                 |
| 1           | $\psi_{X_4}$ |                          |             |                 |
| 2           | $\psi_{X_1}$ |                          |             |                 |
| 2           | $\psi_{X_2}$ |                          |             |                 |
| 2           | $\psi_{X_3}$ |                          |             |                 |
| 2           | $\psi_{Z_1}$ |                          |             |                 |
| 2           | $\psi_{Z_2}$ |                          |             |                 |
| 2           | $\psi_{Z_3}$ |                          |             |                 |

Table 5: Approximate true values of variable importance based on Brier score

| Scenario ID | Variable     | Global survival stacking | Aalen model | Cox-Aalen model |
|-------------|--------------|--------------------------|-------------|-----------------|
| 1           | $\psi_{X_1}$ |                          |             |                 |
| 1           | $\psi_{X_2}$ |                          |             |                 |
| 1           | $\psi_{X_3}$ |                          |             |                 |
| 1           | $\psi_{X_4}$ |                          |             |                 |
| 2           | $\psi_{X_1}$ |                          |             |                 |

Table 5: Approximate true values of variable importance based on Brier score

| Scenario ID | Variable     | Global survival stacking | Aalen model | Cox-Aalen model |
|-------------|--------------|--------------------------|-------------|-----------------|
| 2           | $\psi_{X_2}$ |                          |             |                 |
| 2           | $\psi_{X_3}$ |                          |             |                 |
| 2           | $\psi_{Z_1}$ |                          |             |                 |
| 2           | $\psi_{Z_2}$ |                          |             |                 |
| 2           | $\psi_{Z_3}$ |                          |             |                 |

## References